

Ladera Ranch Did Not Test Soils for Arsenic Trioxide

Here Is How We Know — And What Remains Unchecked

LEHRP independent research note · prepared 2026-07-26 · hypothesis-neutral, no causation asserted

Bottom line. Across every environmental review located and read for the Ladera Ranch master-plan footprint and its immediately adjacent Ranch Plan lands — five original reviews plus two large documents newly full-text searched for this note — none tested soil for **arsenic trioxide**, the chemical historically used in cattle dipping. Only one review tested for arsenic in any form, and it did so for an unrelated reason (former orchard land), using a method that cannot distinguish arsenic trioxide from any other form of arsenic. This is the honest state of the reviewed record — it is not a claim that no document anywhere could ever contradict it. Several named gaps remain, disclosed in full below.

Confidence grades used in this note

Grade	Meaning
A1	Official machine-readable government dataset; peer-reviewed primary research
A2	Official government document, filing, or database, read first-hand
Confirmed	Document fully read/searched by this project
Unread / unlocated	Record known to exist, or believed to exist, but not yet accessed or searched

Every environmental review examined for the Ladera Ranch footprint

Review	Year	Type	Sampled soil?	Arsenic tested?	Grade
Michael Brandman Phase I ESA (Antonio Pkwy EIR App. H)	1995	Phase I (records + visual walk)	No — Phase I by definition	No	A2
Oso Grande School Phase 1 + Addendum (EnviroStor 30020004)	2002–03	Phase I	No	No	A2, confirmed
GeoTracker — all Ladera Ranch cases	—	database check	0 cleanup/investigation cases (2 routine gas-station tank permits only)	—	A2
TTM 17325 / Case 11IC014 (Coto de Caza, adjacent tract, former shotgun range)	2011–15	Phase I -> II -> Remedial Action Plan	Yes, once — the former shotgun range	Lead + PAH only — arsenic not in pages read	A2, confirmed

Review	Year	Type	Sampled soil?	Arsenic tested?	Grade
Case 94IC011 (Coto de Caza Ltd, golf/residential tracts, closed 1998)	1998	OC Local Oversight Program case	Unknown — pre-digital file	Unknown	A2, unread
RanchPlan Affordable Housing Final PEIR (SCH 2015051062)	2016	Program EIR, 1,737 pp	Incorporates FEIR 589 Phase I ESAs by reference	No — 0 hits, full-text searched 2026-07-26	A2, confirmed
FEIR 589 Phase I ESA Appendix I (Planning Areas 1–9)	2003–04	Phase I ESAs, 218 pp	Yes, at one unrelated silica-mine tailings site	1 hit — groundwater, unrelated site (see below)	A2, confirmed
1995 Ladera Planned Community Program Text	1995/2003	likely zoning ordinance	Unknown	Unknown — scanned PDF, no extractable text	not searchable this pass

Verdict on the master-plan footprint itself: “No Action Required” (Oso Grande, 10/7/2003) rests on a paper review alone — no soil was ever sampled. **Verdict on the one place soil WAS sampled nearby (TTM 17325):** tested for the wrong source. When a hazard was already documented — a former shooting range — the regulatory machinery worked as designed: Phase II, a Remedial Action Plan, supervised closure. The 1907–1912 arsenical dip was never entered into any database, so that machinery never engaged. No arsenic-trioxide result exists anywhere in the dip district.

A full-text search of two large documents, done for this note

Two large primary documents were already on file but had not previously been full-text searched. Both were opened and searched programmatically, page by page, for arsenic (and every spelling/spacing variant, including a whitespace-stripped safeguard pass to catch OCR-style extraction artifacts), arsenical, arsenate, cattle dip, dipping vat, tick eradication, quarantine, Texas fever, lead-arsenate, and the organochlorine pesticides (DDT/DDD/DDE, toxaphene, chlordane). Every hit was opened and read in full context before being counted — never reported as a raw keyword-match count.

RanchPlan Affordable Housing Final PEIR (2016), 1,737 pages — zero hits for arsenic in any form, cattle dipping, tick eradication, quarantine, or lead-arsenate. Two hits for the word “trioxide” were read and found to be **“sulfur trioxide” (SO₂)**, an unrelated air-quality/emissions term appearing in a glossary and an air-quality discussion — a genuine false positive from keyword matching, corrected by reading context.

FEIR 589 Phase I ESA Appendix I (2003–04), 218 pages — one genuine “arsenic” hit, unrelated to cattle dipping. Page 119, under Planning Area 5 (“Trampas/Oglebay Norton”) — a silica-sand mine and glass-sand processing plant, a different parcel from the Ladera footprint. A 1999 Phase II investigation of a mining-tailings retention dam examined legacy industrial sand-washing chemicals (sulfuric acid, hydrochloric acid, hydrofluoric acid, sodium hydroxide — none agricultural). Soil trace metals were below EPA residential screening levels; groundwater arsenic was reported above the drinking-water MCL, tested via a standard “Title 22 metals” panel that reports total arsenic, not speciated arsenic trioxide. This is a real, accurate finding — not a false positive — but it concerns groundwater, not soil; an unrelated industrial site, not cattle dipping; and a different planning area, not the Ladera Ranch footprint.

One technical nuance that matters

Even at the few nearby sites that DID test for arsenic (Carl Hankey Elementary, San Juan Elementary, Plant Depot School Site — all outside the Ladera Ranch footprint, tested because of former orchard/agricultural use, not any dipping history), standard environmental laboratory methods report **total elemental arsenic**, not arsenic **speciation**. “Arsenic trioxide” as a distinct chemical species has never been tested for at any site in this record, dip-related or not. Any distinction drawn between the cattle-dip signature (arsenic trioxide, low lead) and the orchard-pesticide signature (lead-arsenate, arsenic bound with lead) is an **inference** from the arsenic-to-lead ratio in results, not a direct laboratory speciation result. That is a real analytical gap even at the sites that did test arsenic.

What has NOT yet been checked — named, disclosed limitations

1. The Ladera Ranch master entitlement EIR itself

County records reference an “EIR 555,” circa 1997, for the original Ladera Planned Community entitlement. Its SCH number and any soil-testing appendix have never been located as a document this project holds. A 1995 “Ladera Planned Community Program Text” IS on file, but it is a **scanned image PDF with zero machine-readable text** (confirmed: 0 of 136 pages yielded any extractable text). No OCR toolchain (tesseract or equivalent) was available in this research pass, so this document could not be searched — it is a disclosed gap, not a null result. By its title it is also more likely the zoning ordinance than the environmental review itself.

2. Case 94IC011 (Coto de Caza, closed 1998)

The file predates digital record-keeping; the online index lists no contaminants or media. What, if anything, was tested is unknown. A public-records request is drafted but has not been sent.

3. TTM 17325 laboratory appendices (“BulkFile1/BulkFile2”)

The narrative pages of this case were read in full, but it was never confirmed whether arsenic was included in the metals suite run alongside lead at the former shotgun range.

4. California State Veterinarian biennial reports (1906–1912)

These large historical documents (100–160 MB each) are held on file. Only specific, already-cited pages have been closely reviewed to date — a full programmatic search of their complete text, of the kind performed on the 2016 PEIR for this note, has not yet been run.

5. Physical, non-digitized archives

A regional historical library, county archives, the California State Archives (Department of Agriculture record groups), and a nature sanctuary that may hold ranch-era photographs or ledgers. Records requests are drafted but not yet sent; none of this material can be searched remotely.

6. State environmental-database per-school profile pages

Automated access to individual school profile pages was blocked during research; only one school's letter has been confirmed read first-hand. For every other school in the district, “no record found” means the question was never opened — not that the result was confirmed clean.

Net assessment. This project holds and has now searched the large majority of the digitized environmental record bearing on this question — including, newly, the two largest documents on file. What remains unchecked is concentrated in three places: one likely-unreadable scanned document (a technical, fixable limitation), two pre-digital county case files, and physical archives that require an in-person visit. None of these are large bodies of untested digital record we are simply choosing not to look at — they are specific, named, and already queued.

This platform is an independent research and data-organization project. It does not provide medical advice and does not establish that any pesticide, property, organization, employer, school, water provider, government agency, or other party caused any illness. Publicly reported health events may not have been independently medically verified. Geographic and temporal overlap does not establish exposure or causation. Formal conclusions require authorized epidemiological analysis, verified medical information, exposure assessment, toxicological review, and independent scientific evaluation.